SAFETY SIGNAL REVIEW

Aggregate safety assessment report

Training project: EdVyro Pharmacovigilance Internship – Safety Signal Review Capstone

Dataset: adverse_drug_reactions_sample.csv (synthetic training dataset; 8 reports)

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1. Executive summary

The dataset contains **8 individual reports** involving 8 different suspected medicines. Three reports (37.5%) record hospitalisation and therefore meet a seriousness criterion on the information supplied. One report is classified as life-threatening; it is the same warfarin report that is also recorded as hospitalised, so it is counted once among serious reports. Three outcomes are recorded as recovered, one as recovering, three as hospitalised, and one as not recovered.

The most prominent individual safety concern for narrative review is the warfarin report ADR-9007 describing uncontrolled haemorrhage, classified as life-threatening with hospitalisation. This is a clinically important event, but a single report in a synthetic dataset cannot establish a new signal or causality. Haemorrhage is a recognised concern with anticoagulant therapy; the report needs clinical context, including INR, dose, concomitant medicines and underlying risk factors.

No reporting dates or exposure denominators are supplied. Therefore, temporal reporting trends, incidence rates, reporting rates, disproportionality statistics, and comparative risk cannot be calculated.

2. Dataset and methods

The CSV has 8 rows and 10 fields: ReportID, PatientAge, PatientGender, SuspectedDrug, Indication, AdverseReaction, ReactionOnsetDays, Severity, Outcome and ReporterType. Descriptive counts and percentages were calculated using the eight reports as the denominator. Percentages are descriptive only and must not be interpreted as incidence or risk because the number of exposed patients and reporting completeness are unknown.

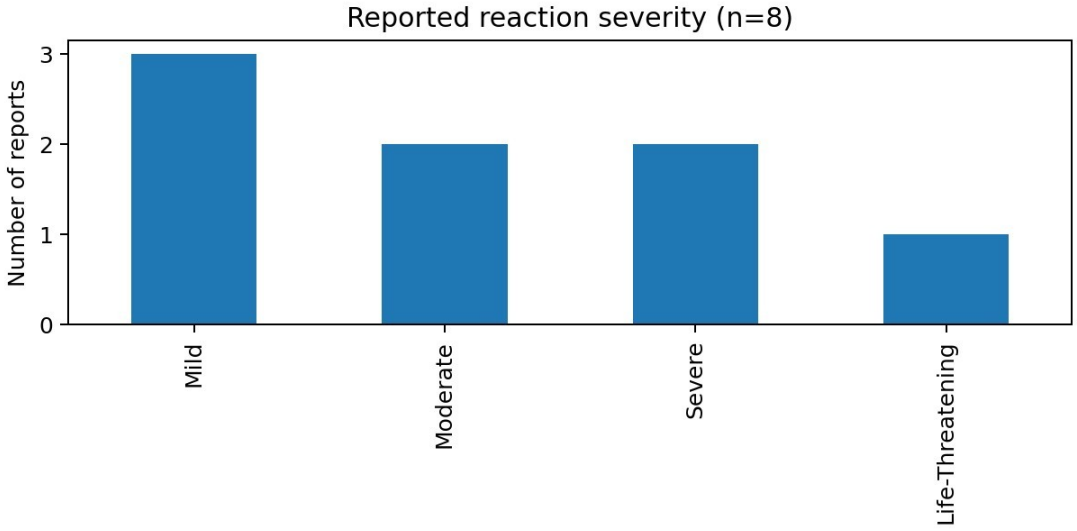
Seriousness was assessed using the regulatory outcome criterion visible in the data: hospitalisation. The life-threatening severity label was also reviewed, but not added as a second case where it occurred in the same report. Other seriousness criteria (death, disability, congenital anomaly, or other medically important condition) cannot be fully assessed from the limited fields.

3. Descriptive findings

Measure	Finding
Reports	8
Unique suspected drugs	8
Age range	28–71 years
Mean / median age	52.0 / 51.5 years
Reaction onset range	1–30 days
Mean / median onset	11.25 / 8.5 days
Gender distribution	Female 4 (50%); Male 4 (50%)

Severity (intensity)

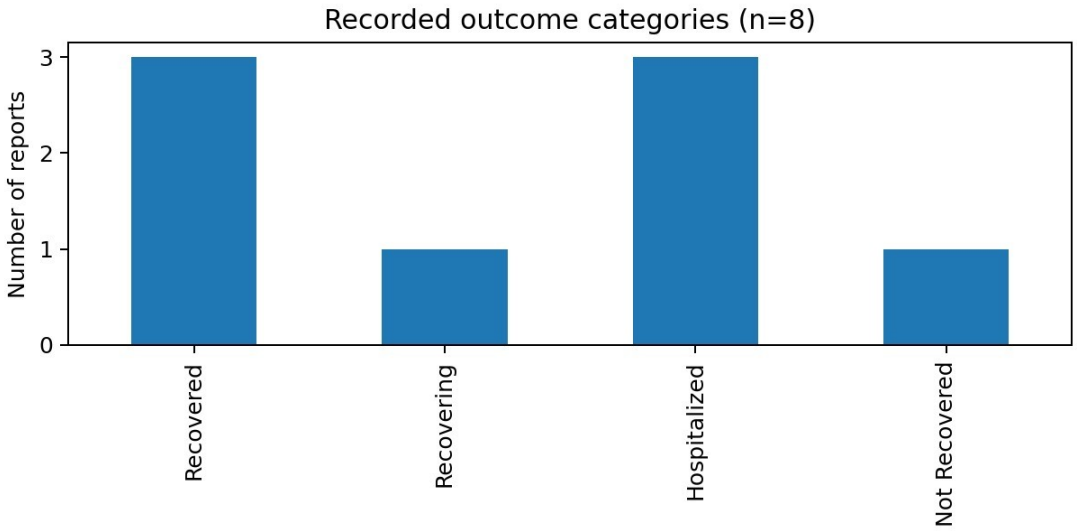
Severity	Reports	Percent
Mild	3	37.5%
Moderate	2	25.0%
Severe	2	25.0%
Life-Threatening	1	12.5%
Total	8	100%



Outcome categories as recorded

Outcome	Reports	Percent
Recovered	3	37.5%
Recovering	1	12.5%
Hospitalized	3	37.5%
Not Recovered	1	12.5%
Total	8	100%

Reporter type	Reports	Percent
Physician	3	37.5%
Pharmacist	2	25.0%
Patient	2	25.0%



Reporter profile

Reporter type	Reports	Percent
Oncologist	1	12.5%

4. Seriousness assessment

Three unique reports record hospitalisation: ADR-9003 (ciprofloxacin; tendonitis), ADR-9004 (ibuprofen; gastric ulcer bleed), and ADR-9007 (warfarin; uncontrolled haemorrhage). These are at least 3/8 (37.5%) serious reports based on the hospitalisation criterion. ADR-9007 is also labelled life-threatening, but remains one report. No death is recorded. The dataset does not contain enough information to determine whether any non-hospitalised report meets another serious criterion.

Severity and seriousness are different: severity describes intensity; seriousness is a regulatory classification based on outcomes/criteria. A severe reaction is not automatically a serious case.

5. Potential safety concern and evidence grading

Case selected: ADR-9007 – warfarin / uncontrolled haemorrhage. The event is labelled life-threatening and the outcome field records hospitalisation. This is clinically important, but a single report in a synthetic dataset cannot establish a new signal or causality. Haemorrhage is a recognised concern with anticoagulant therapy.

Evidence dimension	Assessment
Clinical importance	High: life-threatening haemorrhage and hospitalisation are recorded.
Case count / consistency	Very limited: one warfarin report; no replication or case series.
Temporality	Onset recorded as 10 days; treatment start date and detailed chronology absent.
Clinical plausibility	Plausible generally for anticoagulant therapy; INR and context missing.
Causality	Unassessable from aggregate fields; no structured causality assessment or adequate narrative.
Signal conclusion	Safety concern for follow-up; insufficient evidence to confirm a new signal.

This is best described as a **potential safety concern for follow-up**, not a confirmed or newly detected signal.

6. Alternative explanations and limitations

Possible contributors include excessive anticoagulation/elevated INR; dosing error or recent dose change; interacting prescription, OTC or herbal medicines; underlying gastrointestinal or other bleeding lesion; renal/hepatic dysfunction or other comorbidity; age-related vulnerability; and incomplete reporting. These are hypotheses, not facts established by the dataset.

Key limitations: only eight synthetic reports; one report per suspected drug; no report dates; no exposure denominator; no dose, route, duration or treatment start dates; no concomitant medication list; no INR or other laboratory values; no detailed medical history; no dechallenge/rechallenge information; no full narratives; no reference safety information for each medicine; and no duplicate assessment.

7. Follow-up evidence and risk-minimisation questions

Follow-up area	Questions / evidence to obtain
Case validity and chronology	Confirm patient, reporter, suspected product and event; exact start/stop dates, dose, route, indication and onset.
Bleeding assessment	Bleeding site, clinical severity, treatment, transfusion/procedures, admission/discharge and final outcome.
Anticoagulation monitoring	INR values before/during event; target INR; recent trend and relevant laboratory results.
Concomitant factors	Prescription/OTC/herbal medicines, interaction risks, dosing errors and adherence.
Medical history	Prior bleeding, ulcer disease, liver/renal impairment, malignancy, trauma and comorbidities.
Causality / evidence	Dechallenge, alternative causes, source records and independent clinical review.
Risk minimisation	Were monitoring and counselling followed? Are INR monitoring, interaction checks and bleeding warning signs cleared

8. Conclusions and recommendations

The dataset describes eight reports across eight suspected medicines. Three reports meet the hospitalisation seriousness criterion; one of these is also classified as life-threatening. The warfarin-associated uncontrolled haemorrhage report warrants priority follow-up. However, the evidence is insufficient to infer causality, quantify risk, or confirm a new safety signal.

9. Data integrity note

No reporting dates are present, so temporal trends cannot be analysed. No exposure denominator is present, so incidence or reporting rates cannot be estimated. Percentages describe only this eight-report sample. The CSV is synthetic training material and must not be used for clinical decision-making or regulatory submission.

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